

Bayes's formula

1. Breast cancer is relatively rare among women aged 40–50. About 1% of women in their 40s have breast cancer. Many authorities recommend breast-cancer screening with mammograms starting at age 40. For women in their 40s, mammograms detect about 87% of breast cancers. The “false positive” rate for mammograms given to women in their 40s is about 10%, meaning that about 10% of these women without breast cancer will receive a cancer diagnosis. (In the case of such a diagnosis, physicians follow up with more-invasive tests that have lower false-positive rates.)

Imagine a randomly chosen woman in her 40s. Let B be the event that the woman has breast cancer, and let A be the event that the woman is diagnosed with breast cancer by mammogram.

- a. What is the probability that the randomly chosen woman has breast cancer?
- b. Imagine that the woman has a mammogram and is diagnosed with breast cancer on that basis with breast cancer. Express the probability that the woman has breast cancer given her diagnosis as a conditional probability using A and B . That is the probability that we would like to calculate.
- c. Before calculating, guess the value of the probability from the previous part.
- d. Use the Bayes's formula method to calculate this probability.

2. Two hunters shoot at a deer, which is hit by exactly one bullet. The first hunter hits targets with probability 0.3 and the second hits targets with probability 0.6. The event that the first hunter hits the deer and the event that the second hunter hits the deer are independent. Let's calculate the probability that the second hunter is the one who hit the deer.
 - a. Let B be the event that the second hunter hits the deer. The probability we want is $P(B|A)$ for a certain event A . What is that event?
 - b. Use the Bayes's formula method to calculate $P(B|A)$.